

Comparison of fluorescence-based temperature sensor schemes: Theoretical analysis and experimental validation

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Comparison of fluorescence-based temperature sensor schemes: Theoretical analysis and experimental validation

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The performance of the two most promising fluorescence-based temperature sensing techniques, namely the fluorescence intensity ratio (FIR) and fluorescence lifetime (FL) schemes, have been compared. Theoretical calibration graphs for the two methods illustrate the useful monotonic change of the response with temperature variation. Comparison of the responses and the sensitivities of the two schemes show that at very low temperatures the FIR method exhibits a significant variation with temperature, while the response of the FL method becomes constant with its sensitivity approaching zero. With increasing temperature, the FIR and the FL methods (with short relaxation times and shorter intrinsic lifetimes of the upper energy levels) share a similar sensitivity over a wide temperature range. The presence of a long relaxation time or a longer intrinsic lifetime of the upper level in the use of the FL method gives a less satisfactory response. Experimental data obtained for a range of dopant ions in various host materials are found to be consistent with the theoretical expectation, with each material having a specific energy gap difference. The sensitivities of each material are compared graphically which would allow the most appropriate sensor for an intended application to be selected. © 1998 American Institute of Physics. [S0021-8979(98)04821-X]

INTRODUCTION

Optical fiber temperature sensors have been studied for over two decades¹ and are of particular interest in situations where conventional techniques cannot be used, for example where electromagnetic interference exists. Many optical fiber temperature sensors, however, are also strain sensitive and therefore must be used with care. One solution to this cross-sensitivity problem is to employ the fluorescence from fibers or glass materials doped with rare earth or other similar ions. The underlying physical principle exploited in these temperature sensors is the thermalization which occurs between closely spaced excited ionic energy levels when the levels are populated using an appropriate excitation source, usually a light emitting diode (LED) or a laser diode (LD). The two schemes compared in this paper exploit very different aspects of the resultant fluorescence decay from these levels. Both schemes have the major advantage of being independent of possible fluctuations of the excitation source, which is very important for robust temperature sensing.

Fluorescence lifetime (FL) based temperature sensing techniques have been investigated over a number of years.^{1,2} Pulsed excitation is required and the subsequent temperature-dependent fluorescence decay from the excited levels is monitored with a single detector and appropriate electronics. Analysis of the exponential decay enables the fluorescence lifetime to be determined.

At the same time various investigations into devices based on a fluorescence intensity ratio (FIR) in rare-earth-doped optical material (including fibers) have been undertaken.^{3,4} In this technique the fluorescence from two closely spaced levels to a common final state is monitored, at the appropriate wavelengths, and the ratio of the two emission intensities is calculated, resulting in a temperature-dependent quantity which is independent of the source intensity. Generally a periodic excitation is used in conjunction with phase-sensitive detection to improve the signal-to-noise ratio because the intensity from the upper level will be small at low temperatures.

Both sensor techniques show considerable promise and have been studied in detail for a range of materials, so a comparison of the merits of each scheme is of particular interest. Given the wide range of possible applications for temperature sensors, it is clear that as industry requires specialized sensors for specific processes, no one sensor of either type using a given material or active ion will be suitable for all situations of high or low temperature monitoring, range, sensitivity and stability. However a comparison of the underlying physical basis for the two techniques is of particular value in the development and selection of the most appropriate method to be used in any specific application.

BACKGROUND THEORY

The simplified energy level diagram of a typical fluorescent material depicted in Fig. 1 shows two closely spaced upper states, levels 1 and 2 (energy separation ΔE_{21}), which

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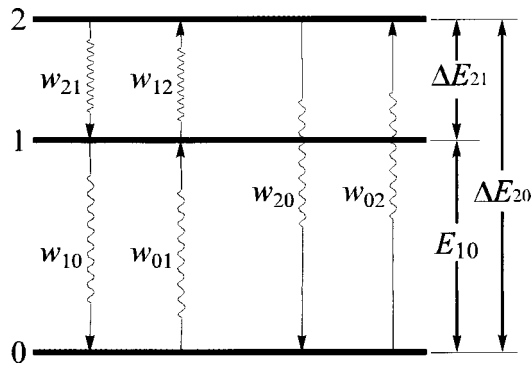


FIG. 1. Schematic energy level diagram of the three level model of a typical fluorescent ion.

may decay radiatively to level 0. When levels 1 and 2 are excited through some pumping scheme (the details of which are not important here) the relevant populations, N_0 , N_1 , and N_2 , respectively, are given by the following rate equations:

$$\begin{aligned} \frac{dN_1}{dt} &= N_2 A_{21} - N_1 (w_{12} + A_{10}) + N_0 w_{01}, \\ \frac{dN_2}{dt} &= N_1 w_{12} - N_2 (A_{21} + A_{20}) + N_0 w_{02}, \\ \frac{dN_0}{dt} &= N_1 A_{10} + N_2 A_{20} - N_0 (w_{01} + w_{02}), \end{aligned} \quad (1)$$

where A_{ij} and w_{ij} are the spontaneous transition rate and the absorption rate, respectively, of the ions from level i to level j (where i, j are integers). Thermalization occurs between the excited levels and so $b_{21} = (g_2/g_1)e^{-\Delta E_{21}/kT}$ represents the ratio of the populations in levels 1 and 2 at equilibrium, where g_1 and g_2 are the respective degeneracies, k is Boltzmann's constant and T is the Kelvin temperature. Applying a Laplace transform to Eq. (1), and if the relaxation time of the upper levels is not very short, the resulting reciprocal lifetime is

$$\frac{1}{\tau} = \frac{1}{2} \left[(1 + b_{21})A_{21} + A_{20} + A_{10} - \sqrt{[(b_{21} - 1)A_{21} + A_{10} - A_{20}]^2 + 4b_{21}A_{21}^2} \right]. \quad (2)$$

When the relaxation time between levels 1 and 2 is extremely short (due to rapid thermalization), the resultant fluorescence lifetime will be given by

$$\tau = (1 + b_{21})/(A_{10} + A_{20}b_{21}), \quad (3)$$

which is the same result as that obtained from the use of a two-level model based on the assumption of thermal equilibrium.^{6,7} Measurements to obtain the lifetime can be performed at either just one of the fluorescence wavelengths or over a larger bandwidth covering all decays from the thermalizing levels. The equations can be readily modified if more than two excited levels are involved.

For radiation at a particular wavelength corresponding to the transition from levels $i \rightarrow f$ the intensity exhibits the proportionality⁷

$$I_{if} \propto N_i \omega_{if} A_{if}, \quad (4)$$

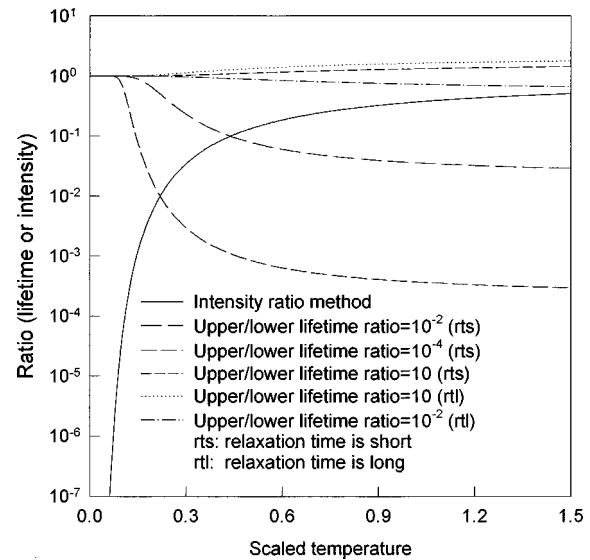


FIG. 2. Comparison of the performance of the FIR and FL methods with scaled temperature variation, for various parameter values.

where ω_{if} is the angular frequency of the radiation. If the separate fluorescence lines from two closely spaced levels (1 and 2 in Fig. 1) are monitored, the intensity ratio for the transitions from levels 1 and 2 to level 0 will therefore be given by

$$\begin{aligned} R = \frac{I_{20}}{I_{10}} &= \frac{N_2 \omega_{20} A_{20}}{N_1 \omega_{10} A_{10}} = \frac{\omega_{20} A_{20} g_2}{\omega_{10} A_{10} g_1} \exp\left(\frac{-\Delta E_{21}}{kT}\right) \\ &= B \exp\left(\frac{-\Delta E_{21}}{kT}\right), \end{aligned} \quad (5)$$

with

$$B = \frac{\omega_{20} A_{20} g_2}{\omega_{10} A_{10} g_1}.$$

The underlying physical basis of the two fluorescence temperature sensors of interest is given in Eqs. (2) and (3) for the FL method and Eq. (5) for the FIR method. It is clear that both techniques result in a quantity which is independent of the source intensity, as required of an effective temperature monitoring scheme, and which is also a nonlinear function of temperature. The actual variation of this quantity with temperature, although very different for the two methods, provides the calibration for the system.

THEORETICAL TEMPERATURE DEPENDENCE AND SENSITIVITY

In order to compare the two methods, the response characteristics given by Eqs. (2), (3) and (5) have been plotted in Fig. 2, using a degeneracy ratio g_2/g_1 of 1, as a function of a scaled dimensionless temperature parameter t , $t = kT/\Delta E_{21}$ (where typical values of $T = 1000$ K, $\Delta E_{21} = 1000 \text{ cm}^{-1}$ give $t = 0.69$). The function represented by Eq. (5) for the FIR method may be plotted directly, but for the FL method it is more convenient to display τ/τ_1 , being the ratio of the thermalized lifetime τ from Eqs. (2) or (3) to the lower level lifetime $\tau_1 (= 1/A_{10})$. The upper and the lower

levels have different lifetimes and so curves for the lifetime method are plotted for three different values of τ_2/τ_1 ($=A_{10}/A_{20}$), where τ_2 is the upper level lifetime, namely 10, 10^{-2} and 10^{-4} and $A_{21} \gg A_{10}$ has been assumed in the plot of Eq. (2). In order to plot Eq. (5) it has been assumed that the transition rate and angular frequency ratios are both 1 (the second assumption is true provided levels 1 and 2 are closely spaced), i.e., $B=1$. Therefore both sets of graphs are shown as ratios on the same vertical axis in which a logarithmic scale has been used in order that the trends may be seen more clearly.

The responses of the two methods to temperature change are very different, even though all the curves exhibit a useful monotonic change in response with temperature variation. At very low temperatures, the fluorescence lifetime is approximately equal to the lifetime of the lower level, and at higher temperatures it either decreases or increases, depending on the intrinsic lifetime ratio of the upper and lower levels. When the intrinsic upper level lifetime is shorter than the lower ($\tau_2/\tau_1 < 1$), the lifetime decreases with temperature increase, but the rate of decrease depends upon the lifetime ratio of the upper and lower levels, as well as the relaxation time. The smaller lifetime ratio of the two levels with the shorter relaxation time [i.e., Eq. (3)] gives a greater variation in the lifetime with temperature. When the upper level intrinsic lifetime is longer than the lower ($\tau_2/\tau_1 > 1$), the lifetime increases as the temperature increases and when the intrinsic lifetime of the upper level is greater the variation with tem-

perature is larger. The intensity ratio, on the other hand, increases rapidly with increasing temperature at low temperature, and the rate of increase decreases with increasing temperature until it asymptotes to 1 at very high temperatures. It is clear that both methods will not work particularly well at very high temperatures (approximately $T > 1000$ K, where fortunately blackbody techniques become relatively easy to use), and that the fluorescence lifetime method also will not perform satisfactorily at very low temperatures ($T < 200$ K). However, even though the intensity ratio varies significantly at low temperatures, such a large difference in intensities may be difficult to detect in a practical system in the presence of system noise. At room temperature either method could be used, as in each case the quantity of interest shows significant variation with temperature.

In comparing these two types of sensors it is also worthwhile to compare the sensitivity S defined as the fractional rate of change of the response r with temperature, i.e., $(1/r)(\partial r/\partial T)$. The sensitivity for the FIR method may be obtained from Eq. (5) as

$$S = \frac{1}{R} \frac{\partial R}{\partial T} = \frac{\Delta E_{21}}{kT^2} = \frac{k}{\Delta E_{21}t^2}, \quad (6)$$

while for the FL method (where S_l is defined as the sensitivity for a long relaxation time and S_s for a short relaxation time) the expression may be written as

$$S_l = \frac{1}{\tau} \frac{\partial \tau}{\partial T} = \frac{-\frac{A_{21}}{A_{10}}b_{21} + \frac{\left(\frac{A_{21}}{A_{10}}b_{21} + \frac{A_{21}}{A_{10}} + 1 - \frac{A_{20}}{A_{10}}\right)\frac{A_{21}}{A_{10}}b_{21}}{\sqrt{\left((b_{21}-1)\frac{A_{21}}{A_{10}} + 1 - \frac{A_{20}}{A_{10}}\right)^2 + 4\left(\frac{A_{21}}{A_{10}}\right)^2b_{21}}} \frac{k}{\Delta E_{21}t^2}}{(1+b_{21})\frac{A_{21}}{A_{10}} + \frac{A_{20}}{A_{10}} + 1 - \sqrt{\left((b_{21}-1)\frac{A_{21}}{A_{10}} + 1 - \frac{A_{20}}{A_{10}}\right)^2 + 4b_{21}\left(\frac{A_{21}}{A_{10}}\right)^2}} \frac{k}{\Delta E_{21}t^2}, \quad (7)$$

$$S_s = \frac{1}{\tau} \frac{\partial \tau}{\partial T} = \left(1 - \frac{1}{1+b_{21}}\right) \left(\frac{1 - \frac{A_{20}}{A_{10}}}{1 + \frac{A_{20}}{A_{10}}b_{21}}\right) \frac{k}{\Delta E_{21}t^2}. \quad (8)$$

In Fig. 3 the magnitude of the sensitivity-energy gap product, $S \cdot \Delta E$, obtainable directly from Eqs. (6)–(8), has been plotted as a function of the scaled temperature, as this gives an expression which is independent of the energy gap. Three different value of the ratio of the upper to lower level lifetimes have been used and the degeneracy ratios have been set to 1, as before. It is interesting to observe the very similar sensitivities for the FIR and FL curves over much of the temperature range when the intrinsic lifetime ratio of the upper to lower level is smaller than 1 and the relaxation time

is very short (such as occurs with alexandrite and verified by Gayen *et al.*,⁸ where the nonradiative relaxation time was 27 ps for the ${}^4T_2 \rightarrow {}^2E$ transition). However, when the relaxation time is long or the upper lifetime is larger than the lower, the sensitivity reduces considerably over the same temperature region for the FL scheme. Among materials for FL data having long relaxation time, the use of an upper/lower ratio larger than 1 gives better sensitivity than its counterparts having a ratio smaller than 1. At very low temperatures, the sensitivity of the FL method diminishes significantly, which is to be expected given the flattening of the corresponding response curves in Fig. 2. In contrast, the FIR sensitivity is enhanced at very low temperatures although, as noted above, this corresponds to a very large intensity ratio and it may be affected significantly by system noise.

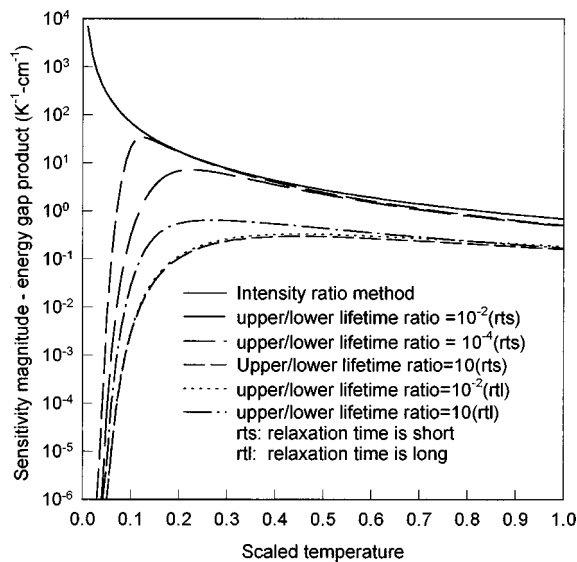


FIG. 3. Sensitivity-energy gap products of the FIR and FL methods as a function of the scaled temperature, for various parameter values.

COMPARISON OF THEORY WITH DATA

In order to verify the above theoretical analysis a number of experimental data sets for the two methods are displayed in Figs. 4 and 5 and compared with theory. Figure 4 shows FL data, displayed as τ_2/τ_1 (as in Fig. 2), plotted as a function of scaled temperature. The FL data were obtained from crystals of ruby,¹ alexandrite,⁹ $\text{Cr}^{3+}:\text{YAG}$,¹ and bulk $\text{Pr}^{3+}:\text{ZBLAN}$ (Ref. 5) and have been fitted to Eq. (3) for the first three materials, as shown in the diagram. In the case of praseodymium Sun *et al.*⁵ found that the relaxation time is of the same order of magnitude as the intrinsic lifetimes of the upper and lower levels and the upper/lower lifetime ratio is approximately 2, which implies that Eq. (2) should be used to fit the data. For ruby, alexandrite and $\text{Cr}^{3+}:\text{YAG}$, some data in the higher temperature region were not used in the

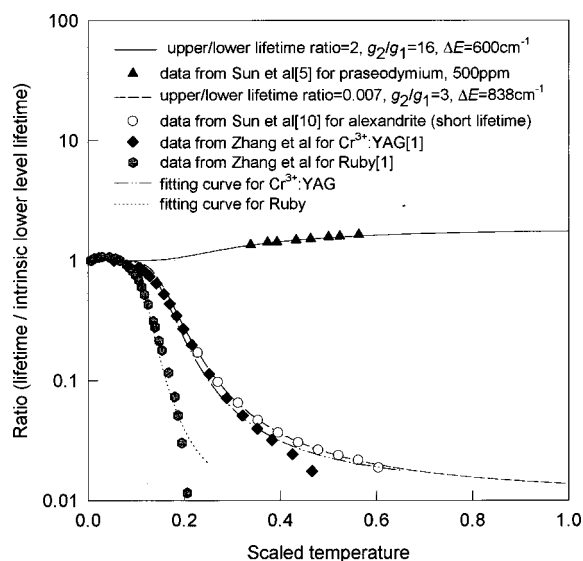


FIG. 4. The lifetime ratio data as a function of the scaled temperature for crystals of ruby, alexandrite, $\text{Cr}^{3+}:\text{YAG}$ and bulk $\text{Pr}^{3+}:\text{ZBLAN}$, where fitting curves are plotted based on Eqs. (3) and (2).

fitting procedure, as the lifetime decreases rapidly as a function of temperature at higher temperatures where thermal quenching occurs—an effect which cannot be described with the simplified model implicit in Eq. (3). Elsewhere¹ it has been shown that these data can be fitted very well when more energy levels than shown in Fig. 1 are included. Nevertheless the fit to data for these four materials is very good and confirms the approach implied in Eqs. (2) and (3).

In Fig. 5 normalized FIR data are shown as a function of the scaled temperature for Er^{3+} -doped fiber,⁴ Yb^{3+} -doped fiber¹⁰ and bulk $\text{Pr}^{3+}:\text{ZBLAN}$.¹¹ Each data set was fitted using Eq. (5) in which the normalizing factor B and ΔE_{21} were the free parameters. All data were divided by the appropriate value of B , and so the three data sets in Fig. 5 follow the same curve, as anticipated from Fig. 2. The agreement between the experimental results and theory is again very good and the simple theory of operation of FIR sensors implied by Eq. (5) is confirmed.

TEMPERATURE RESPONSES AND SENSITIVITIES

On the basis of the theory behind the FL and FIR responses, it is now appropriate to compare the non-normalized data with fitted curves as functions of temperature so that actual measurement requirements may be appreciated. The system responses in Figs. 6 and 7 (data from Figs. 4 and 5) show the lifetime and intensity ratio for the FL and FIR methods, respectively, with Eqs. (6), (7), and (8) used to fit the data.

Praseodymium is one of the few cases where data are available for both methods using similar samples over comparable temperature ranges. It can be seen that the variation of the FIR method is more rapid than that of the FL method over the temperature region from 250 to 550 K. The signal-to-noise ratio is assumed to be high so that system noise is not an important consideration.

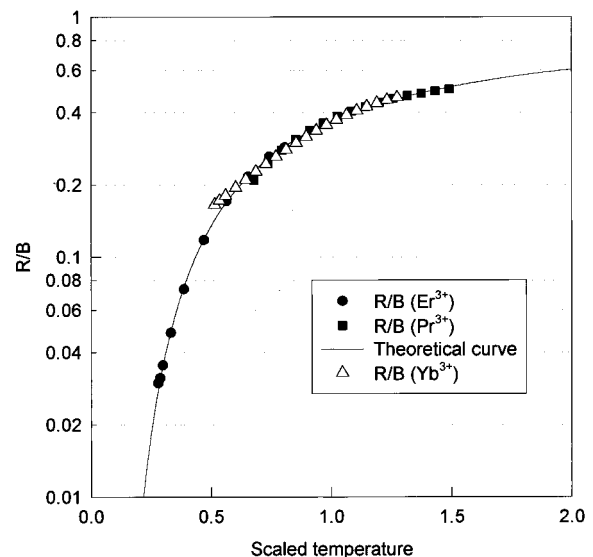


FIG. 5. Normalized FIR data as a function of the scaled temperature for Er^{3+} -doped fiber, Yb^{3+} -doped fiber and bulk $\text{Pr}^{3+}:\text{ZBLAN}$, where fitting curves are plotted based on Eq. (5).

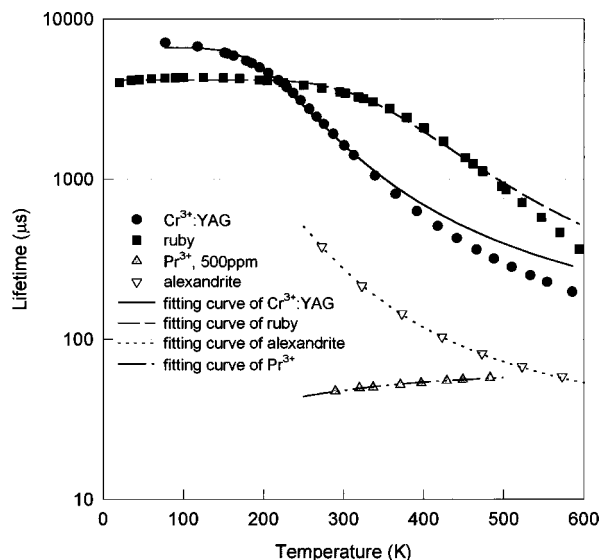


FIG. 6. The temperature responses for the FL scheme for crystals of ruby, alexandrite, Cr^{3+} :YAG and bulk Pr^{3+} :ZBLAN, where fitting curves are plotted based on Eqs. (2) and (3).

For the other FL data the relaxation time between the upper and lower levels is extremely short and the resultant lifetime, described by Eq. (3), is greater. Alexandrite^{1,9} is a typical example and the lifetime ratio (for the short lifetime) matches the theoretical curve very well using the following parameters: τ_2/τ_1 is approximately 0.007, a degeneracy ratio of 3 and energy gap between 4T_2 and 2E is assumed⁹ to be 838 cm^{-1} . As seen in Fig. 4, a larger difference between the two intrinsic lifetimes and the shorter lifetime of the upper level gives a greater variation of lifetime with temperature.

The duration of the fluorescence lifetime as well as its sensitivity with temperature should be considered when designing a FL sensor. If the lifetime is too short, the frequency response of the detector arrangement should be chosen to

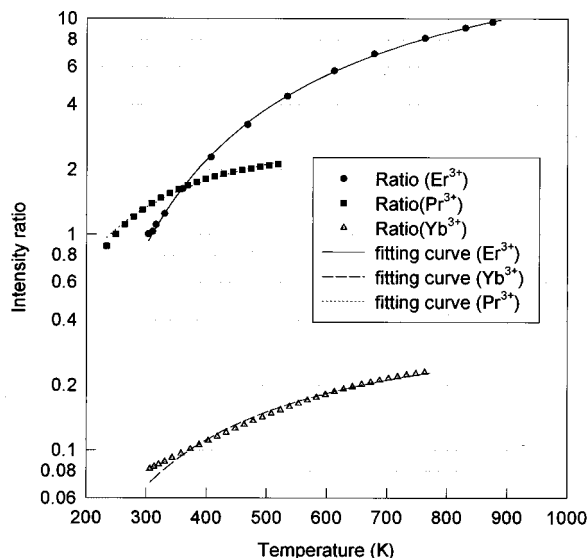


FIG. 7. The temperature responses for the FIR scheme using Er^{3+} -doped fiber, Yb^{3+} -doped fiber and bulk Pr^{3+} :ZBLAN, where fitting curves are plotted based on Eq. (5).

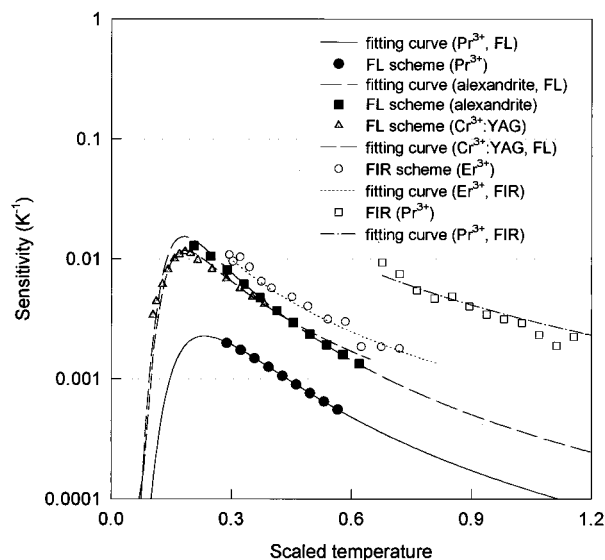


FIG. 8. A global comparison of the sensitivities of both methods, comparing experimental and theoretical data.

match the signal, while the existence of a long lifetime makes the measuring procedure longer in a practical instrument based on this approach, which is not ideal for real-time monitoring.

The data in Fig. 7 for the FIR method in three dopant materials have intensity ratios not too far removed from 1, which should enable any of them to be used in practical devices over the indicated temperature ranges with a good signal-to-noise ratio.

Figure 8 shows a comparison of the sensitivities of both methods by using all the data previously presented and the theoretical curves fitted to the data. The best sensitivity is that of FIR with Pr^{3+} , over the temperature region of 250–500 K, supporting the theory described above. For the FL scheme, the sensitivity of alexandrite is better than that of praseodymium and Cr^{3+} :YAG, and for the FIR scheme, the sensitivity of Pr^{3+} is better than that of Er^{3+} . The wide variety of sensitivities reported show that this important parameter is not only closely related to the upper/lower lifetime ratio and the relaxation time, as shown clearly in Fig. 3, but also to the energy gap difference. For the same material, Pr^{3+} :ZBLAN glass, the sensitivity of the FIR scheme is greater than that of the FL scheme over the temperature range considered, which supports the theory that the sensitivity of a material with a longer relaxation time is less than when using the intensity ratio method (FIR), for the same material.

DISCUSSION

The above comparison shows that at very low temperatures the FIR method has a higher sensitivity and is thus preferred for temperature monitoring because the response for the FL method tends to a constant value with its sensitivity approaching zero. However, over a wider range the difference between the responses is less evident, especially in the case where the intrinsic lifetime of the upper level is shorter than the lower and the relaxation time is short. Many

other factors need to be taken into account in any practical comparison of these responses, including the spectroscopic details of the dopant ion and the technical issues associated with constructing a sensor based on these results. The behavior of such an instrument based on either method will also depend on which levels of the particular dopant ion are involved. In particular the signal-to-noise ratio actually achieved will be strongly related to factors such as the source used, its wavelength and its relationship to the absorption spectrum, the ease of coupling to the optical fiber, the detection sensitivity and the geometry of the probe. Thus very large intensity ratios may cause problems of accuracy, as do very short lifetime measurements ($\tau \leq 1 \mu\text{s}$) in low cost instrumentation. However, the analysis presented here is helpful in the selection, and a process of cross referencing and cross calibration of one method to the other may be used to widen and optimize the range of the sensor, in the same way that the use of blackbody radiation has been demonstrated as a cross-referencing approach to measurement of both higher ($T > 500 \text{ K}$) and lower ($T < 500 \text{ K}$) temperatures with the same probe.¹²

The presence of the blackbody radiation can present a problem with the intensity ratio method at higher temperatures ($T > 500 \text{ K}$) unless it is effectively filtered out, or the wavelength of measurement is short ($\lambda < 700 \text{ nm}$) where the effect is less obvious. However, at $T > 1000 \text{ K}$ it is almost impossible to remove unless a pulsed approach to the fluorescence excitation is used, enabling discrimination against the direct current blackbody radiation background which is inevitably present. Usually for the FL method, when the relaxation time is relatively short, a shorter intrinsic lifetime of the upper level occurs and a higher sensitivity can be obtained for measurement purposes. However, when the relax-

ation time is long, a material with a longer upper level lifetime gives a better sensing characteristic. Both the FIR and FL methods with a short relaxation time and a shorter upper level lifetime, share a similar sensitivity over a wide range of temperatures despite having a different response to temperature variations.

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